

Step-by-Step Implementation Documentation (Submission Item 2)

Objective

To deploy a working PrestaShop e-commerce store on AWS Free Tier using:

- 1) **Amazon EC2** as the web/application server
- 2) **Amazon RDS MySQL** as a **separate database** (not installed on the EC2 instance)

Implementation Order:

I created the **EC2 instance first**, then created the **RDS database**, and connected them during the PrestaShop installation.

Step 1: Create the EC2 Application Server (First)

- 1) Log into the AWS Console as root user → EC2 → **Launch instance**
- 2) Name: Input the instance name as *prestashop-app*
- 3) AMI Selection: **Ubuntu Server 22.04 LTS**
- 4) Instance type: Select the **t3.micro** (Free Tier eligible)
- 5) Key pair: Create a new key pair and download the *.pem* file
- 6) Network settings:
 - I. Create a new security group
 - II. Allowed **SSH (port 22)** from my IP address only
 - III. Allowed **HTTP (port 80)** from 0.0.0.0/0
 - IV. Allowed **HTTPS (port 443)** from 0.0.0.0/0 (optional)
 - V. Launched the instance in **eu-north-1** region

Configuration Decision: Ubuntu 22.04 LTS + t3.micro was chosen because it provides excellent support for PHP 8.3+ required by PrestaShop 9.1 and stays within Free Tier limits.

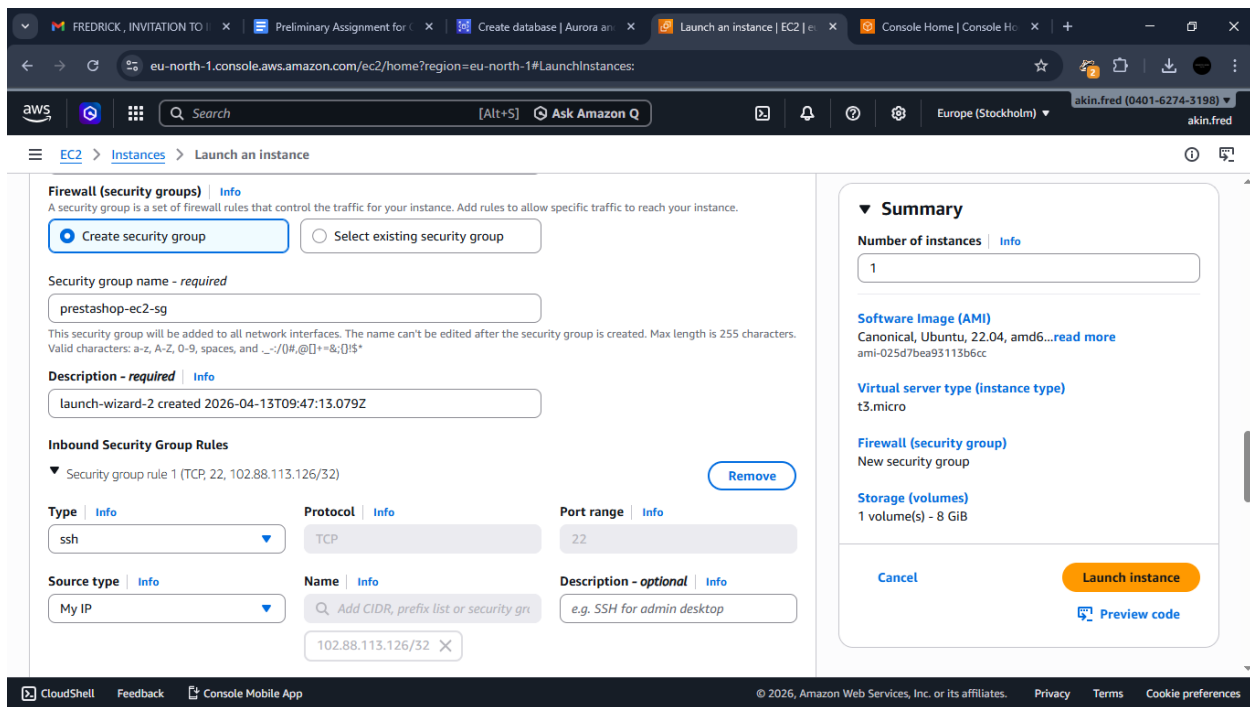


Figure 1 EC2 “Launch instance” summary page

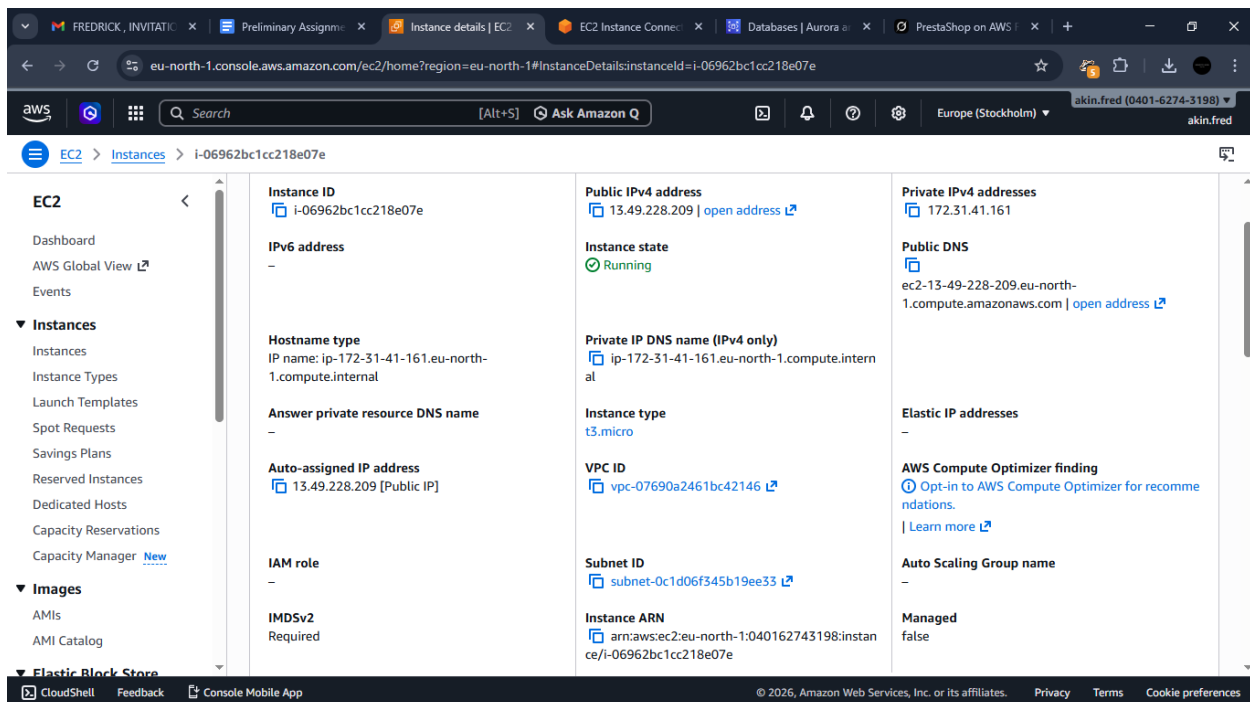


Figure 2 Instance in “Running” state with Public IPv4 DNS: `ec2-13-49-228-209.eu-north-1.compute.amazonaws.com`

Step 2: Create the RDS Database (Second – Separate from EC2)

- 1) From the AWS Console, search RDS → Aurora & RDS → **Create database**
- 2) From the Engine options: Select **MySQL** (latest version)
- 3) Template: **Free tier**
- 4) DB instance identifier: prestashop-db
- 5) Master username: admin
- 6) Master password: [strong password set]
- 7) Instance class: **db.t3.micro**
- 8) Storage: 20 GiB gp2
- 9) Connectivity: **Publicly accessible: Yes** (for this assignment only)
- 10) Additional configuration → Initial database name: left blank (I created it manually later)
- 11) Created the database in the same **eu-north-1** region

Configuration Decision: RDS was created separately after the EC2 instance to ensure the database runs on a dedicated managed MySQL server and not on the application server (EC2). This meets the assignment requirement for separate hosting.

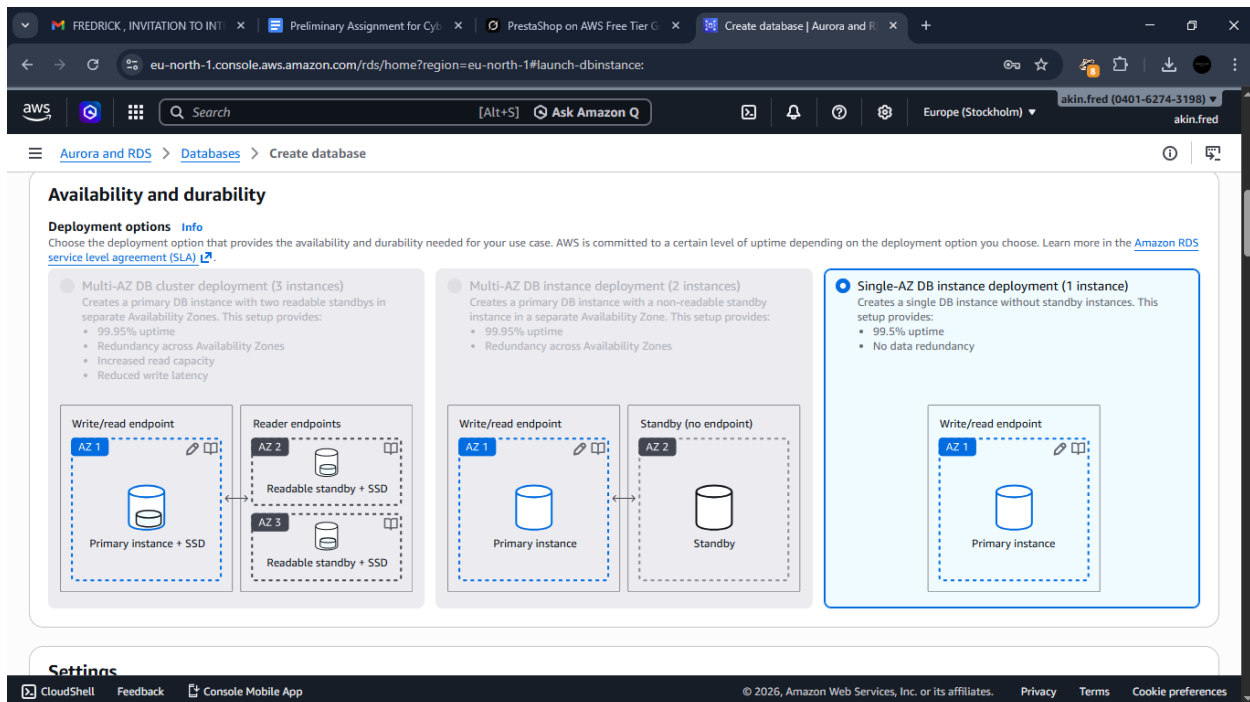


Figure 3 RDS Database Creation page

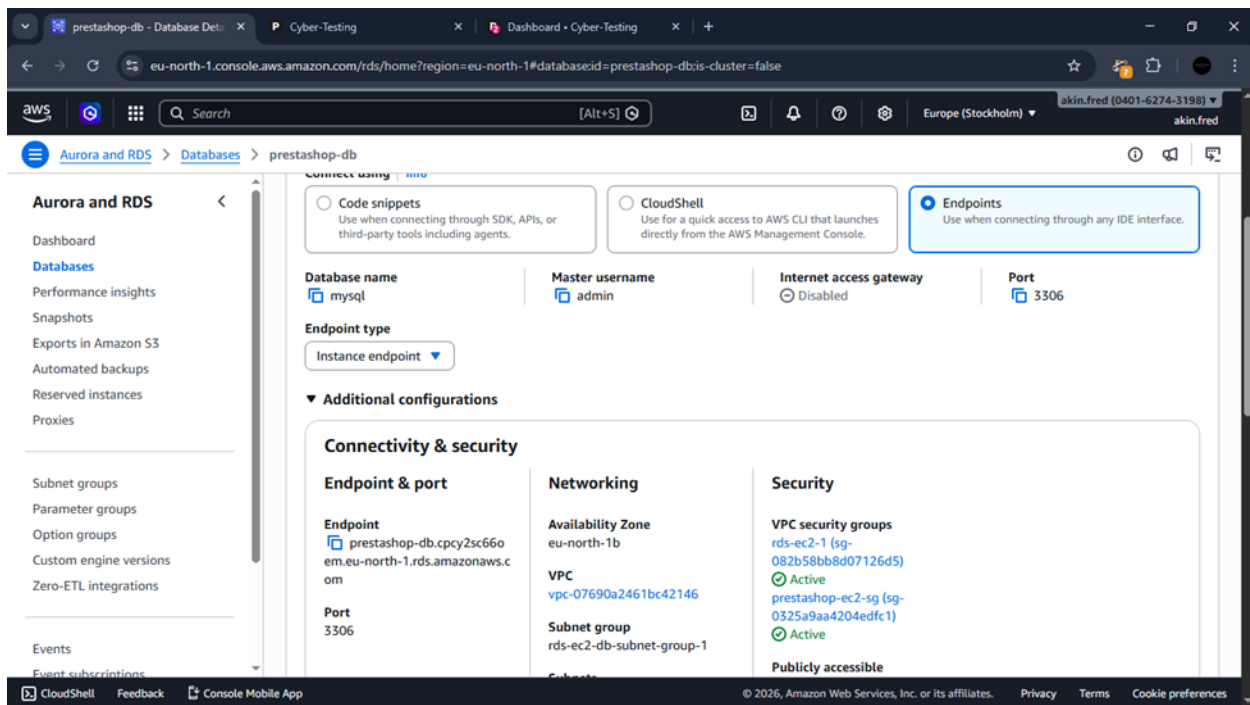


Figure 4 RDS Database Availability status

Step 3: Set Up the LAMP Stack and PrestaShop on EC2

- 1) Connected to EC2 via SSH using the key pair
- 2) Open the Command-line interface on the host PC (CLI) and navigate to the storage location of the downloaded key pair (I used GIT BASH.) and run the following commands:
 - `“chmod 400 prestashop-key.pem”` It sets the file permissions to read-only for the owner and no permissions for anyone else (group or others)
 - `“ssh -I prestashop-key ubuntu@ ec2-13-49-228-209.eu-north-1.compute.amazonaws.com”`. To connect to a remote server using SSH with a specific private key.
- 3) Updated system and installed Apache, PHP 8.3 and required extensions using the following commands on the command line:
 - `sudo apt update && sudo apt upgrade -y`
 - `sudo apt install apache2 php8.3 php8.3-mysql php8.3-gd php8.3-curl php8.3-xml php8.3-mbstring php8.3-zip php8.3-intl php8.3-opcache unzip -y`
 - `sudo systemctl enable --now apache2`
- 4) Download the official Prestashop release and extract it using the following commands:
 - `cd /var/www/html`
 - `sudo wget https://api.prestashop-project.org/assets/prestashop-classic/9.1.0-4.0/prestashop.zip -O prestashop.zip`

- sudo unzip prestashop.zip

5) Set proper file permissions

- sudo chown -R www-data:www-data /var/www/html
- sudo chmod -R 755 /var/www/html

```
ubuntu@ip-172-31-41-161:/var/www/html$
Warning: prestashop_edition_basic_version_9.1.0-4.0.zip?token=9c3095b0a0:
Warning: Permission denied
0 117M 0 1360 0 0 4843 0 7:02:53 --:--:-- 7:02:53 4839
curl: (23) Failure writing output to destination
ubuntu@ip-172-31-41-161:/var/www/html$ ls
index.php
ubuntu@ip-172-31-41-161:/var/www/html$ curl -O https://assets.prestashop3.com/ds
t/edition/corporate/9.1.0-4.0/prestashop_edition_basic_version_9.1.0-4.0.zip?tok
en=9c3095b0a0
% Total % Received % Xferd Average Speed Time Time Time Current
Dload Upload Total Spent Left Speed
0 0 0 0 0 0 0 0 --:--:-- --:--:-- --:--:-- 0w
Warning: Failed to create the file
Warning: prestashop_edition_basic_version_9.1.0-4.0.zip?token=9c3095b0a0:
Warning: Permission denied
0 117M 0 1360 0 0 6896 0 4:56:59 --:--:-- 4:56:59 6903
curl: (23) Failure writing output to destination
ubuntu@ip-172-31-41-161:/var/www/html$ ubuntu@ip-172-31-41-161:/var/www/html$ curl -O https://assets.prestashop3.com/dst/edition/corporate/9.1.0-4.0/prestashop_edition_basic_version_9.1.0-4.0.
zip?token=9c3095b0a0
% Total % Received % Xferd Average Speed Time Time Time Current
Dload Upload Total Spent Left Speed
0 0 0 0 0 0 0 0 --:--:-- --:--:-- --:--:-- 0w
Warning: Failed to create the file
Warning: prestashop_edition_basic_version_9.1.0-4.0.zip?token=9c3095b0a0:
Warning: Permission denied
0 117M 0 1360 0 0 6896 0 4:56:59 --:--:-- 4:56:59 6903
curl: (23) Failure writing output to destination
^C
ubuntu@ip-172-31-41-161:/var/www/html$ ^C
ubuntu@ip-172-31-41-161:/var/www/html$ sudo curl -O "https://assets.prestashop3.com/dst/edition/corporate/9.1.0-4.0/prestashop_edition_basic_version_9.1.0-4.0.zip?token=9c3095b0a0"
% Total % Received % Xferd Average Speed Time Time Time Current
Dload Upload Total Spent Left Speed
100 117M 100 117M 0 0 81.2M 0 0:00:01 0:00:01 --:--:-- 81.2M
ubuntu@ip-172-31-41-161:/var/www/html$ ls
index.php prestashop_edition_basic_version_9.1.0-4.0.zip?token=9c3095b0a0
ubuntu@ip-172-31-41-161:/var/www/html$ ls -la
total 120956
drwxr-xr-x 2 www-data www-data 4096 Apr 13 10:44 .
drwxr-xr-x 3 root root 4096 Apr 13 10:32 ..
-rwxr-xr-x 1 www-data www-data 960007 Mar 23 11:02 index.php
-rw-r--r-- 1 root root 122885233 Apr 13 10:44 'prestashop_edition_basic_version_9.1.0-4.0.zip?token=9c3095b0a0'
ubuntu@ip-172-31-41-161:/var/www/html$ sudo mv 'prestashop_edition_basic_version_9.1.0-4.0.zip?token=9c3095b0a0' prestashop.zip
ubuntu@ip-172-31-41-161:/var/www/html$ sudo chown www-data:www-data prestashop.zip
ubuntu@ip-172-31-41-161:/var/www/html$ sudo chmod 644 prestashop.zip
ubuntu@ip-172-31-41-161:/var/www/html$ sudo systemctl restart apache2
ubuntu@ip-172-31-41-161:/var/www/html$ ls -la
total 1900
drwxr-xr-x 2 www-data www-data 4096 Apr 13 10:46 .
drwxr-xr-x 3 root root 4096 Apr 13 10:32 ..
-rw-r--r-- 1 www-data www-data 2998 Apr 13 10:45 XMLFeed.cache
-rwxr-xr-x 1 www-data www-data 960007 Mar 23 11:02 index.php
-rw-r--r-- 1 www-data www-data 960007 Apr 13 10:46 index.php.temp
-rw-r--r-- 1 www-data www-data 4429 Mar 23 11:02 prestashop.zip
ubuntu@ip-172-31-41-161:/var/www/html$
```

Figure 5 Terminal showing successful installation of LAMP stack and PrestaShop files

Step 4: Run the PrestaShop Web Installer & Connect to RDS

- 1) Open browser and went to: <http://ec2-13-49-228-209.eu-north-1.compute.amazonaws.com>
- 2) Followed the installer:
 - Store information → Fill the form with the correct and necessary details.
 - **Database configuration** (this is where EC2 and RDS were joined):
 - Database server address: prestashop-db.cpcy2sc66oem.eu-north-1.rds.amazonaws.com
 - Database name: prestashop-db
 - Database login: admin
 - Database password: [password used]

- Tables prefix: ps_

Issue Encountered & Resolution: Received error: “Connection to MySQL server succeeded, but database 'prestashop' not found”. **Solution:** Connected to RDS from EC2 using MySQL client and manually ran:

```
CREATE DATABASE prestashop-db CHARACTER SET utf8mb4 COLLATE utf8mb4_general_ci;  
GRANT ALL PRIVILEGES ON `prestashop-db`. * TO 'admin'('@'%);  
FLUSH PRIVILEGES;
```

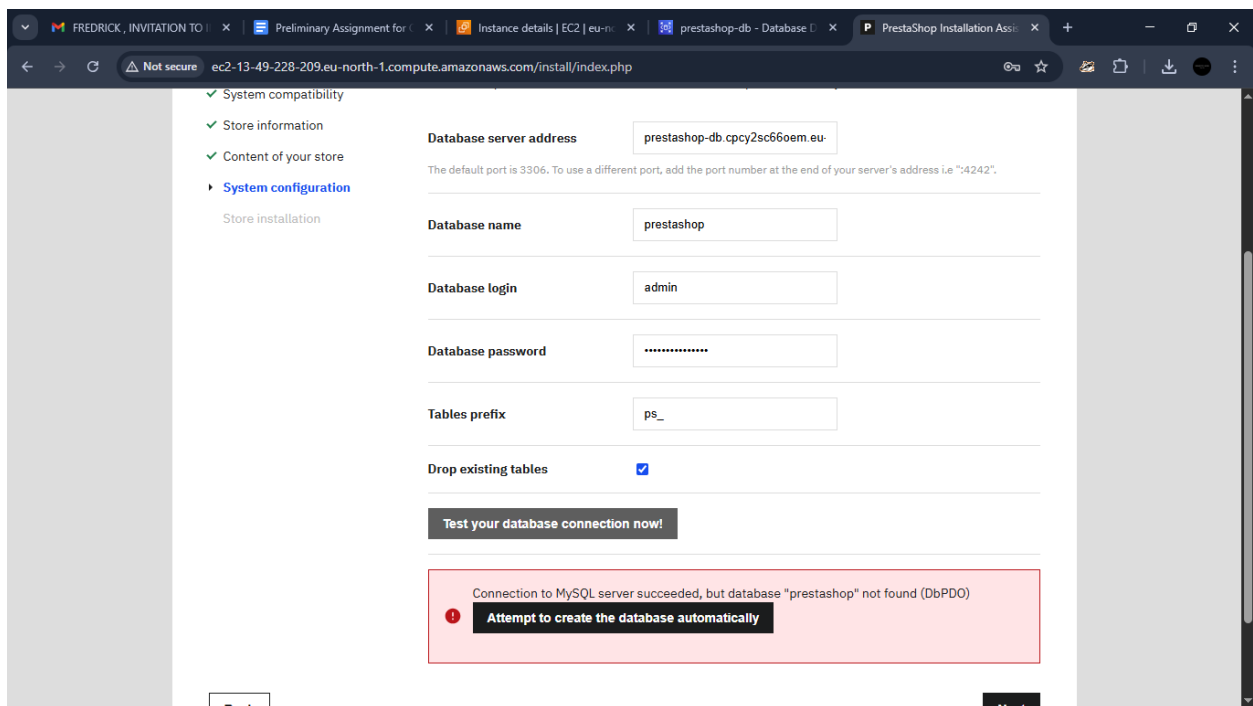


Figure 6 Database connection error

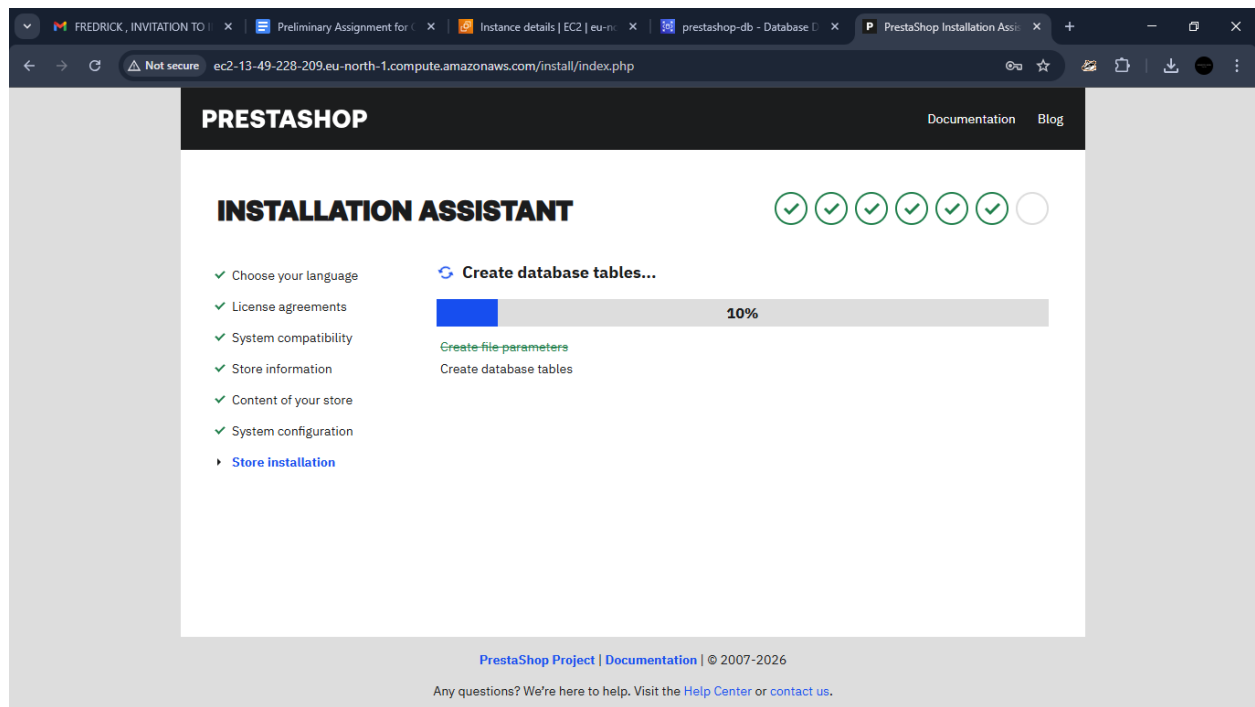


Figure 7 Database connection successful and table creation

Step 5: Post-Installation Security Step

1) Deleted the /install folder for security:

- *Bash*
- *sudo rm -rf install*

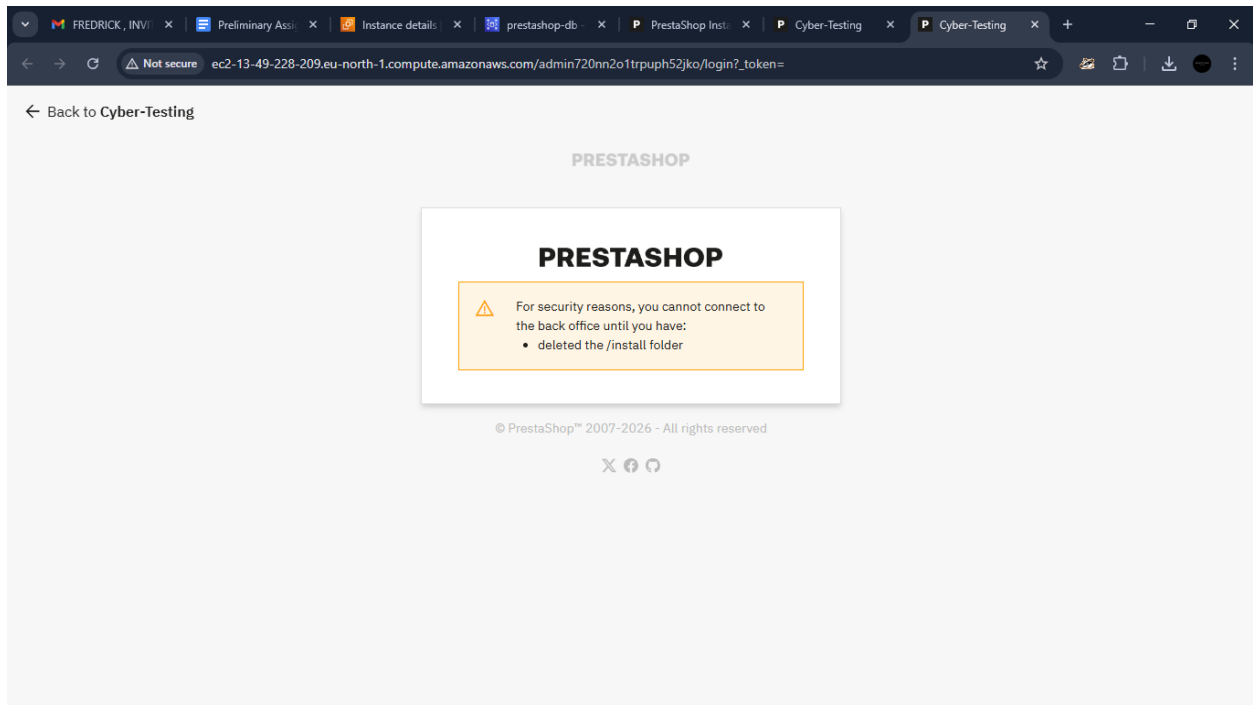


Figure 8 Security warning page requiring deletion of /install folder

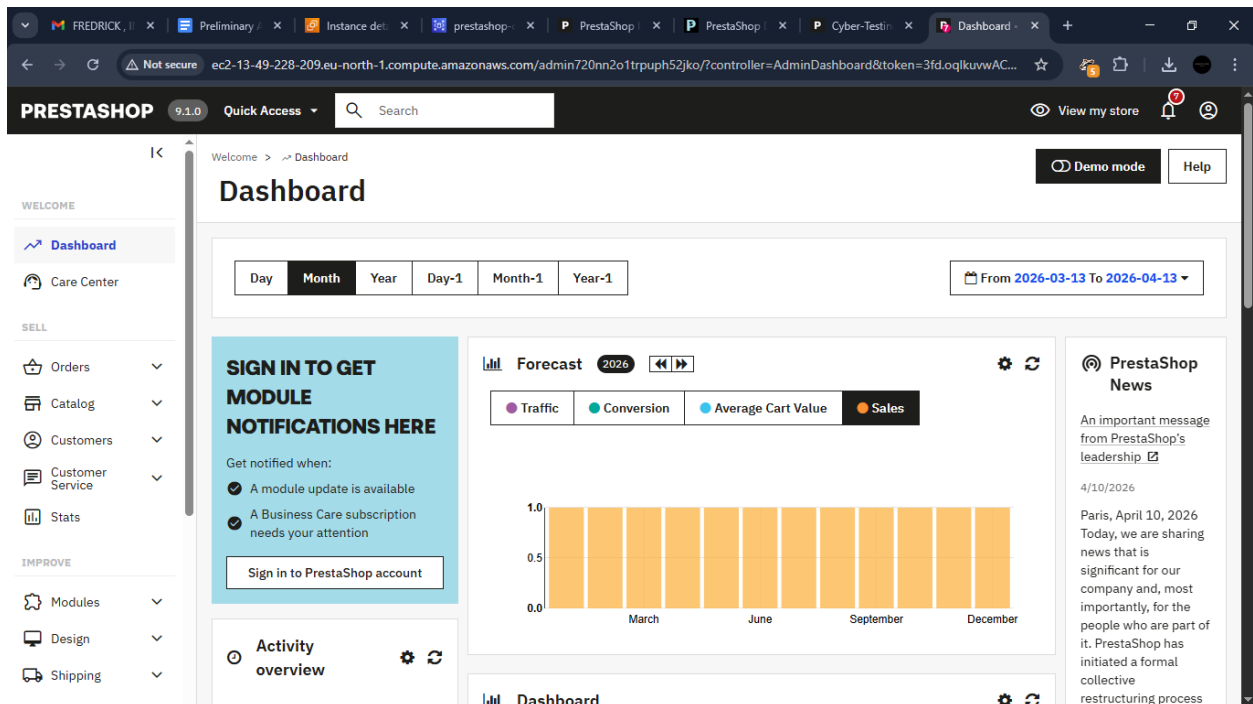


Figure 9 Back Office login page after deleting the /install folder

Step 6: Final Verification

- 1) Live store is accessible and functional
- 2) Database is hosted on dedicated RDS instance (separate from EC2)
- 3) All resources stayed within Free Tier limits

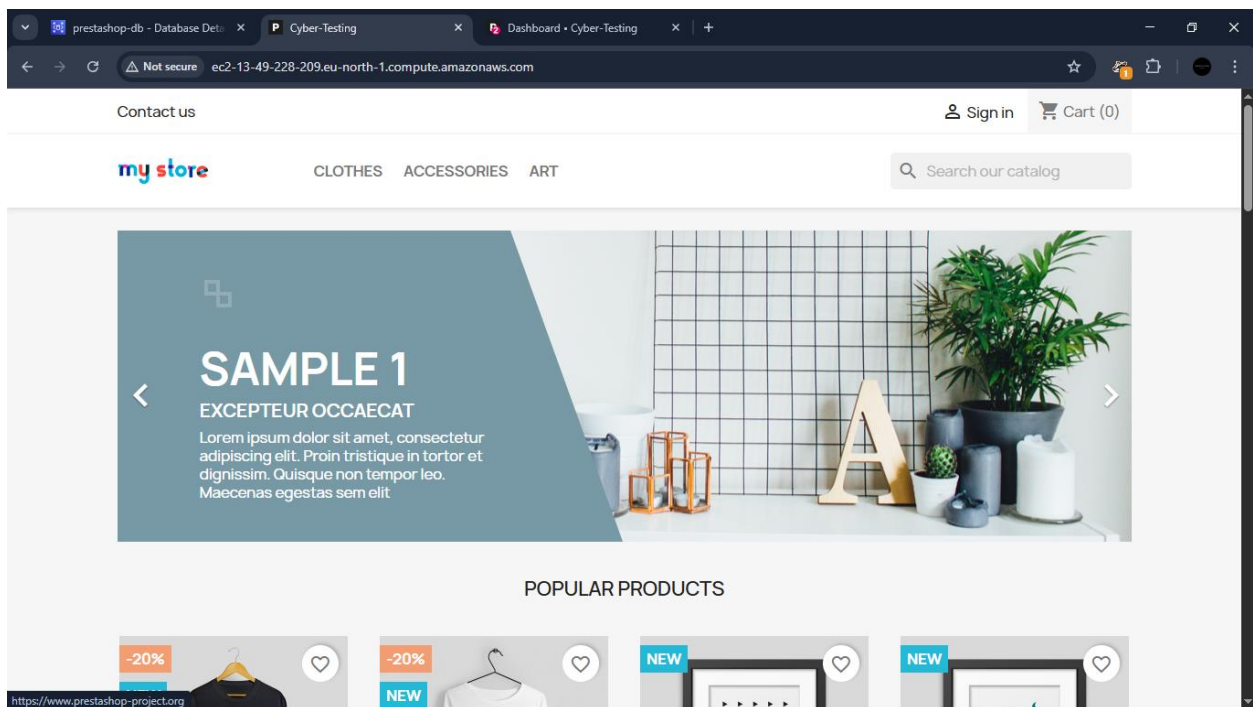


Figure 10 Live PrestaShop homepage showing Cyber-Testing store

Conclusion

I successfully deployed PrestaShop 9.1 by first launching the EC2 instance, then creating the RDS database, and connecting them during installation. The store is now live and publicly accessible.